

**WEB-BASED MANAGEMENT SYSTEM WITH PREDICTIVE ANALYTICS FOR
PORT MANAGEMENT UNIT OF PASACAO**

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CHAPTER I

THE PROJECT AND ITS BACKGROUND

Introduction

Technology plays a vital role in improving operations across various sectors, including ports and fisheries management offices. Through the use of advanced systems, many essential processes can now be performed more efficiently, accurately, and with minimal manual effort. An Information System is defined as a system that collects, processes, stores, analyzes, and distributes information for a specific purpose (Rainer & Prince, 2021). In recent years, many Information Systems have incorporated predictive analytics, which utilizes historical data, statistical techniques, and machine learning methods to forecast future trends and outcomes (Kumar et al., 2021). These technological advancements enable organizations to enhance performance by ensuring accurate data recording, efficient report generation, and improved decision-making. In fisheries and port management, predictive analytics can assist administrators in identifying patterns in revenues and estimating future operational trends based on historical data.

This study will be conducted at the Port Management Unit (PMU) located in Zone 2, Barangay San Cirilo, Pasacao, Camarines Sur. The PMU was established 21 years ago and is currently managed by Mr. Jan Benedict P. Fabi, who serves as the designated Officer-in-Charge (OIC). The PMU is responsible for supervising fish port operations in the municipality and serves as the primary landing site for

fishery products delivered by local fishermen and commercial fishing vessels, with approximately 1,400 fishermen. The office manages the recording of transactions and collection of fees such as Facility Usage Fees (rental, entrance, parking, storage, usage, and wharfage), Fish Unloading Fees, Auxiliary invoice Fee, and Accreditation Fee, as part of its daily operations. These responsibilities highlight the importance of maintaining accurate and organized operational records. Therefore, efficient data management is essential to support the functions and overall performance of the unit.

At present, the Port Management Unit (PMU) relies on manual processes in recording operational and financial transaction information derived from various port activities such as facility usage (rental, entrance, parking, storage, usage, and wharfage), fish unloading, auxiliary invoice, and accreditation. Although multiple types of transactions are conducted daily, the recording and consolidation of their corresponding data are handled manually and maintained in separate files. As a result, data is not centrally stored, making the process time-consuming and prone to human error. As transaction volume increases, organizing, verifying, and summarizing data for report generation becomes more difficult, leading to delays and inconsistencies in records. The absence of a centralized and structured system further affects data accuracy, as transaction information may be incomplete, inconsistently formatted, or improperly recorded. In addition, the PMU does not maintain an organized inventory of operational resources, making it difficult to monitor and manage materials and equipment needed for port activities. Furthermore, the PMU does not utilize any form of data analytics to identify

revenue trends, limiting its ability to support data-driven decision-making. As a result, the PMU experiences difficulty in planning and allocating budgets for port operational needs.

To address these issues, the researchers propose the development of a Web-Based Fish Port Operations Management System with Revenue Forecasting for the Port Management Unit (PMU) of Pasacao. The proposed system will record and store all transaction data in a centralized database, ensuring accurate and consistent records through automated fee computation. It will also include an operational resource inventory management module to monitor and manage materials, equipment, and supplies used in port activities. Additionally, the system will generate daily, monthly, and annual reports for efficient monitoring and evaluation. Moreover, it will incorporate predictive analytics to analyze historical transaction data and weather conditions to forecast future revenue trends, providing valuable insights for decision-making, particularly in planning, budgeting, and staffing. Through these features, the system aims to enhance overall operational efficiency, accuracy, and data management within the PMU.

Objectives

The main objective of this study is to design and implement a Web-Based Fish Port Operations Management System with Revenue Forecasting for the Port Management Unit (PMU) of Pasacao to support transaction recording, inventory management, and revenue forecasting. Specifically, this study aims to:

1. Develop a transaction management system for fish port operations.

2. Develop an operational resource inventory management module.
3. Develop a predictive analytics module that uses forecasting models to estimate future revenue.

Scope and Delimitations

This study involves the development of a web-based fish port operations management system with revenue forecasting for the Port Management Unit (PMU) of Pasacao, Camarines Sur. The system will be accessible only to authorized personnel, specifically the port manager and the statistician, and will implement role-based access control to ensure that each user has designated permissions and interfaces. The port manager will oversee and monitor overall fish port operations, review operational and financial analysis through the system dashboard, including revenue forecasts, and support decision-making within the PMU. The statistician, on the other hand, will be responsible for encoding, updating, and managing operational and financial transaction records within the system.

The transaction management component of the system is limited to handling and organizing transaction information only. It does not cover the actual physical transaction processes occurring in the port. Physical transactions, such as fish unloading, fee assessment, and collection, are carried out manually by different personnel due to limited access to digital devices. These transaction details are then consolidated and forwarded to the statistician, who records them in the system. The recorded data will serve as the basis for report generation and

submission to the municipal government. The system therefore functions as a centralized digital repository of transaction records rather than a real-time transaction processing system.

The system will allow the statistician to record and manage transaction data and stakeholder profiles, such as renters, buyers, and brokers. The recorded data will serve as the basis for generating reports such as daily collection monitoring, monthly collection monitoring, and annual collection monitoring, as required by the Port Management Unit. These reports will also be used for submission to the Mayor's Office and the Municipal Treasury Office for monitoring and documentation purposes. However, system usage is limited to office-based operations, as only the statistician and the port manager have access to devices capable of running the system.

The system will include an operational resource inventory module that allows the statistician to record, update, and monitor port resources such as equipment, materials, and supplies. It will store basic information including item name, quantity, and status to support inventory tracking. The port manager will use the inventory data and forecasted revenue to support budgeting and resource planning, especially during peak and off-peak seasons. However, the system is limited to recording and monitoring inventory data only and does not include automated procurement, supplier management, or real-time tracking of physical inventory movement.

The forecasting process will utilize historical transaction data and weather conditions, as these factors significantly influence fishing activities and overall port revenue. Forecasts will be generated twice a year: January to June, which represents the peak season, and July to December, which represents the off-peak season. This approach will help the port manager plan operations more effectively and support budget preparation for the Local Government Unit (LGU). The generated forecasts will be displayed through the system dashboard for the port manager to assist in planning and decision-making processes.

Furthermore, the accuracy of the predictive analytics module may affect the accuracy because of the quality and completeness of the available data. Historical records, previously stored in Excel spreadsheets and manual logbooks, may contain inconsistencies, missing entries, or data transfer errors. These limitations may affect the reliability of the forecasts generated by the SARIMA model. Therefore, the results should be interpreted as estimates intended to support planning and informed decision-making, rather than exact or guaranteed predictions.

This study is limited to the internal operations of the Port Management Unit (PMU) and is designed as an operational support tool. The system will be web-based and accessible only to authorized personnel. It will not include features such as online payment processing, mobile application development, hardware integration, or integration with external government systems. The system is not intended to replace official accounting, auditing, or financial management systems, but rather to support data organization and decision-making.

Significance of the Study

This study, which involves the development of a Web-Based Fish Port Operations Management System with Revenue Forecasting for the Port Management Unit (PMU) of Pasacao, will be significant to the following:

Port Management Unit of Pasacao. The implementation of the proposed system will improve the efficiency of transaction recording, fee computation, profiles of stakeholders, and report generation. It will also assist management in monitoring inventory and forecasting revenue trends to support operational planning and decision-making.

Port Personnel. The system will reduce the workload associated with manual record-keeping, minimize errors in documentation, and allow faster retrieval of transaction records.

Local Government Unit (LGU). The system will provide more accurate and organized operational and collection reports, which can support monitoring and policy development related to fisheries management.

Programmers. This study will serve as a reference for programmers in developing web-based management systems that integrate transaction management, cloud storage, data analysis, and predictive analytics.

Researchers. This study will serve as a reference material for future research related to operations management systems and predictive analytics in fisheries and port management.

Definition of Terms

Inventory. A detailed list of all the items in a place or the process of counting and recording goods and materials (Cambridge Dictionary, 2026). In this study, it refers to the operational resources such as materials, equipment, and supplies used in port activities.

Operational Resources. The assets, personnel, and infrastructure essential for carrying out daily operations and achieving organizational goals (WisdomLib, 2026). In this study, it refers to the materials, equipment, and supplies used in fish port operations that are recorded and monitored through the inventory module.

Predictive Analytics. The practice of using statistical algorithms and machine learning techniques to analyze historical data, identify patterns, and predict future outcomes (GeeksforGeeks, 2025). In this study, it is a system feature that analyzes past revenue records to forecast expected income trends.

Revenue. The total income produced by a given source (Merriam-Webster, 2026). In this study, these are the total fees collected by the Port Management Unit (PMU) from recorded transactions.

Seasonal AutoRegressive Integrated Moving Average model (SARIMA) . An extension of the traditional ARIMA model, specifically designed for time series data with seasonal patterns (GeeksforGeeks, 2025). In this study, SARIMA is utilized as the forecasting model to analyze historical data and predict future revenue.

Stakeholders. A person, group, or organization that has an interest in a business or project and can affect or be affected by its activities (GeeksforGeeks, 2025). In this study, these refer to the buyers, brokers, and renters involved in port operations within the Port Management Unit (PMU).

Conceptual Framework

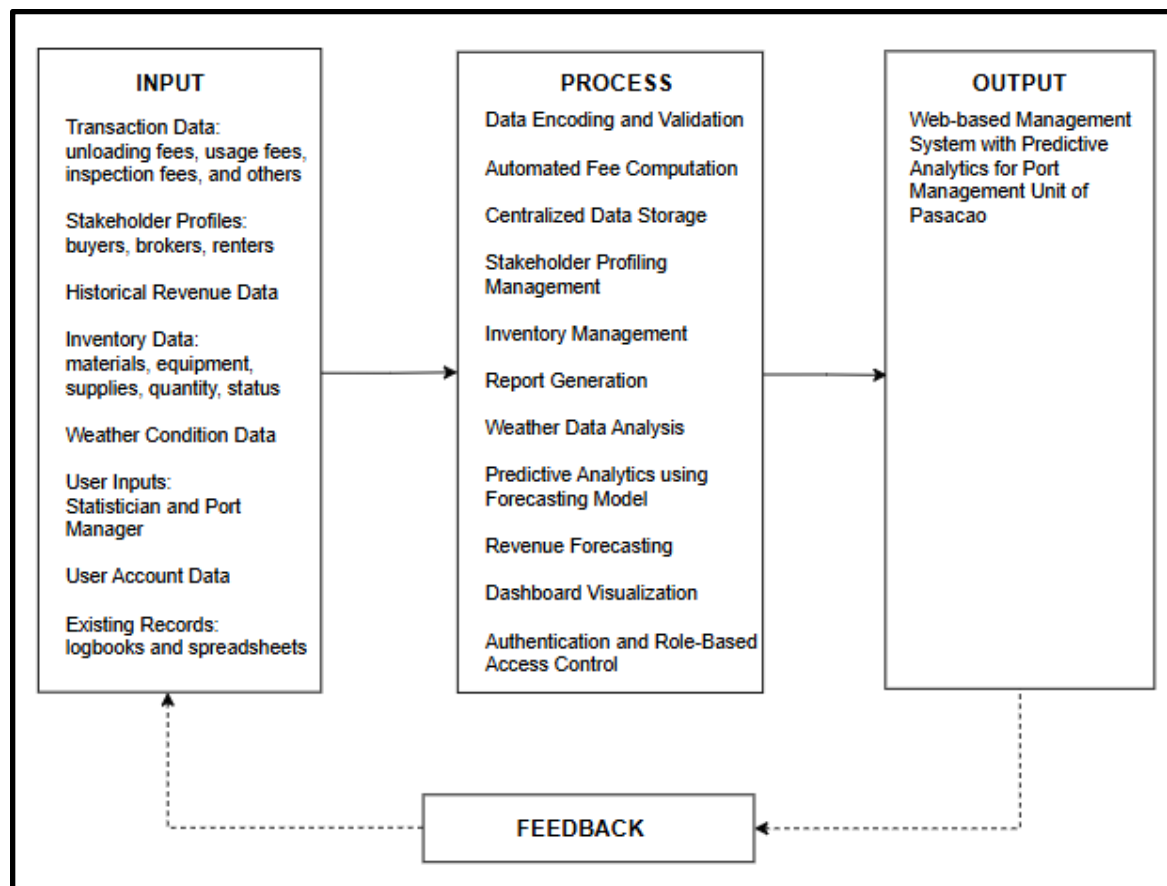


Figure 1
**Conceptual Framework of the Web-based Management System with
Predictive Analytics for Port Management Unit of Pasacao**

Figure 1 illustrates the Conceptual Framework of the Web-Based Management System with Predictive Analytics for Port Management Unit (PMU) of Pasacao using the Input-Process-Output (IPO) model. The framework presents the flow of data within the system, beginning from data collection, followed by data processing, and ending with the generation of useful outputs for monitoring and decision-making. It also includes a feedback mechanism that supports continuous system improvement. Through this structure, the framework highlights how manual and fragmented processes are transformed into a centralized and efficient digital management system.

In the Input Phase, the framework focuses on the collection of necessary data used by the system. This includes Transaction Data such as unloading fees, usage fees, inspection fees, and other related transactions that represent the financial activities within the fish port. Stakeholder Profiles contain information about buyers, brokers, and renters involved in port operations. Historical Revenue Data serves as the basis for revenue analysis and forecasting. Inventory Data includes records of materials, equipment, supplies, quantity, and status used in port operations. Weather Condition Data is also included to support operational monitoring and analysis. In addition, User Inputs from the Statistician and Port Manager allow system interaction and management, while Existing Records from logbooks and spreadsheets are integrated into the system to preserve data

continuity and historical reference. These inputs provide the essential information required for system processing.

The Process Phase represents the transformation of collected data into organized, accurate, and meaningful information. This begins with Data Encoding and Validation to ensure data accuracy and consistency. Automated Fee Computation calculates transaction fees efficiently and minimizes manual errors. Centralized Data Storage organizes all collected information into a structured database for easier access and management. Stakeholder Profiling Management maintains and updates stakeholder records, while Inventory Management monitors the availability and status of port resources and supplies. Report Generation produces daily, monthly, and annual reports for operational monitoring. Weather Data Analysis evaluates weather-related information that may affect port activities. Predictive Analytics using the Forecasting Model analyzes historical revenue data to identify trends and patterns, while Revenue Forecasting generates estimated future revenues to support planning and decision-making. Dashboard Visualization presents summarized data through graphical reports for easier interpretation. Lastly, Authentication and Role-Based Access Control restrict system access to authorized users to ensure data security and proper system management.

In the Output Phase, the framework produces the Web-Based Management System with Predictive Analytics for the Port Management Unit of Pasacao. This output integrates all processed information into a centralized platform that supports real-time monitoring, data management, report generation, and revenue

forecasting. Through the system, authorized users can access organized records, monitor operational activities, analyze revenue trends, and utilize predictive insights for better planning and decision-making. The system improves efficiency, accuracy, accessibility, and overall management of fish port operations.

The framework also includes a Feedback Loop that connects the output back to the input phase. This represents the continuous improvement of the system through updated records, user feedback, and system adjustments. Information generated from reports, monitoring activities, and forecasting results can be used to refine future inputs and improve operational processes. This feedback mechanism ensures that the system remains adaptive, accurate, and responsive to the changing needs of the Port Management Unit (PMU).

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter represents an in-depth review of literature and related studies concerning the development of the Web-Based Fish Port Operations Management System with Revenue Forecasting for the Port Management Unit (PMU) of Pasacao.

Fish Port Operations Management System

The digital transformation of maritime transport has significantly influenced the development of modern port management systems. Digital technologies improve port operations by enhancing communication, information sharing, and coordination among port stakeholders. Smart port systems also improve operational efficiency by implementing digital technologies that allow ports to manage logistics activities and operational processes more effectively. Both ideas highlight that the adoption of digital platforms enables port authorities to process operational data more efficiently and improve the coordination of port services (Tijan et al., 2021; Toro et al., 2024).

Modern port management systems are also supported by advanced technologies that enhance monitoring and operational decision-making. Smart port systems integrate technologies such as automation, the Internet of Things (IoT), and data analytics to improve operational monitoring and logistics coordination. These technologies allow ports to collect and analyze operational data from multiple sources, including vessels, cargo operations, and port infrastructure.

Through data analysis, port authorities can identify inefficiencies and improve the performance of port operations (Zhang et al., 2024).

In addition, port management systems provide structured platforms for recording and monitoring operational activities. These systems allow the recording of operational data such as vessel schedules, cargo operations, and port logistics activities. By organizing operational data in a digital system, port managers can easily monitor daily activities, retrieve historical records, and improve reporting processes (Woldesemayat, 2025).

Recent studies have also emphasized the role of artificial intelligence in improving port operations. Artificial intelligence-based smart port logistics systems are designed to enhance productivity and safety in port environments. These technologies support real-time monitoring and data-driven decision-making in port logistics. AI systems can analyze operational data and identify patterns that help port managers optimize logistics operations and improve operational efficiency (Sim et al., 2024).

Web-Based Operational Management Systems

Web-based operational management systems play an important role in improving how organizations handle their daily activities. These systems help automate common tasks such as data recording, report generation, and monitoring of operations. By reducing manual work, organizations can complete tasks faster and with fewer errors. This also helps employees manage their workload more efficiently and focus on more important tasks. In addition, these systems provide

a platform where data can be stored, managed, and accessed easily, which supports better decision-making (Bueno et al., 2024).

One of the main advantages of web-based systems is their ability to improve data organization and accessibility. These systems store information in a structured and digital format, making it easier to search, retrieve, and update records. By replacing paper-based processes, organizations can avoid delays and reduce the risk of losing important documents. Users can also access data in real time, which improves coordination and communication among staff. This leads to better planning and more effective operational management (Vega-Quintana et al., 2024).

Moreover, web-based systems enhance operational efficiency by improving data accuracy and communication across different units. As part of information systems, they allow users to access updated data anytime and from different locations. This flexibility helps organizations respond quickly to operational needs and unexpected changes. It also improves coordination between departments by providing a shared platform for information exchange. As a result, workflows become more organized and efficient (Ebiesuwa et al., 2023).

In addition, the use of automated data processing and databases improves the reliability of operational systems. Instead of manually recording data, information can be entered and processed automatically within the system. This reduces human errors and ensures that data remains consistent and accurate. It also allows faster handling of large amounts of data, which is important in busy

operational environments. In the Philippines, a web-based information system was developed to digitize operations and improve efficiency, data accuracy, and decision-making in local government units. These improvements help organizations strengthen their monitoring and reporting capabilities (Enaño et al., 2025).

Furthermore, web-based management systems support overall organizational performance by improving productivity and reducing operational costs. Automation minimizes repetitive tasks and reduces the need for excessive manual labor. At the same time, real-time data access allows managers to make timely and informed decisions. These systems also improve transparency by providing clear and updated records of operations. In addition, web-based systems support planning and decision-making by allowing users to analyze and use data effectively in managing operations. Studies have shown that such systems enhance data-driven decision-making by providing accessible and organized information for users. Because of these benefits, web-based systems are considered essential tools for modern organizations aiming to improve efficiency and performance (Mercurio & Hernandez, 2022; Lapada, 2023).

Transaction Management System

Advancements in digital technology have significantly transformed maritime and trade management through the integration of web-based information systems and automated data processing. In the fish supply chain, traditional management of landing transactions through handwritten ledgers is highly inefficient, as

manually recorded data in logbooks often leads to incorrect writing and inevitable data entry errors. This shows that manual systems are prone to inaccuracies and delays. In port operations, where data is time-sensitive, these errors can lead to incorrect revenue collection and reporting. Because of this, improving the method of recording transactions is necessary (Lim & Arbaiy, 2022).

The use of digital systems improves the accuracy and speed of transaction recording by allowing data to be processed and stored in real time. These systems reduce manual workload and minimize errors in recording transactions. Digital platforms also provide more organized and reliable records that can be easily accessed when needed. In the Philippines, digital systems have been shown to improve data accuracy, timeliness, and transparency by replacing manual processes. As a result, transitioning from manual to digital transaction management systems improves efficiency, accuracy, and reliability in port operations (Santos & Reyes, 2023).

Beyond simple data entry, recent literature emphasizes the importance of securing digital records to maintain operational integrity. Digitalizing maritime logistics not only improves speed but also enables a paperless audit trail that is instantly accessible to administrative staff. However, as systems shift to digital platforms, the need for strong security becomes more important. Implementing Role-Based Access Control (RBAC) helps protect sensitive port data by restricting system access to authorized personnel only. This ensures that only assigned users can modify records and maintain data integrity. Because of this, data remains

accurate, secure, and protected in digital systems (Paulauskas et al., 2021; Ahasan, 2024; Fortinet, 2026).

Inventory Management for Port Operations

Inventory management systems play a crucial role in improving operational efficiency and resource control in organizations. Automated inventory systems enhance data accuracy and minimize manual errors while improving the tracking and monitoring of resources. These systems also organize inventory data in a structured manner, allowing faster retrieval and more efficient management of materials, equipment, and supplies. In addition, real-time tracking supports timely updates and helps reduce data redundancy, leading to better decision-making. In contrast, manual inventory processes often result in delays, inconsistencies, and difficulties in maintaining accurate records, which negatively affect operational performance (Odasco & Saong, 2023; Gumilao et al., 2024; Alhaddad et al., 2024).

In port environments, inventory management is closely related to resource monitoring and logistics operations. Inventory and resource management are essential in port operations, as they ensure the availability and proper handling of materials, equipment, and supplies required for daily activities. Modern port systems use digital platforms to monitor and manage operational resources, allowing better tracking and control of inventory-related processes. These systems improve data accuracy and enable real-time monitoring, which helps prevent shortages and operational delays. By organizing resource data into centralized systems, port managers can make informed decisions regarding resource

allocation and utilization. These advancements show that digital port systems support inventory management functions such as tracking, monitoring, and controlling operational resources in port environments (Paraskevas et al., 2024).

Predictive Analytics Using SARIMA

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model has been widely applied in financial and operational forecasting due to its ability to capture both trend and seasonal patterns in time-series data. Studies have shown that SARIMA is particularly effective in analyzing historical financial datasets, allowing organizations to identify recurring patterns and generate accurate forecasts. In business applications, SARIMA has been used to predict profit and revenue trends, demonstrating its capability to handle seasonal fluctuations and support data-driven decision-making. These findings suggest that SARIMA is a reliable method for forecasting revenue-related data, making it applicable to systems such as fish port operations where financial transactions exhibit time-dependent and seasonal behavior (Sirisha et al., 2022; WDSI, 2022).

In addition, SARIMA is useful for large-scale forecasting and planning. Research on mining revenue found that SARIMA can accurately predict long-term trends and seasonal changes, helping improve decision-making and resource allocation. Similarly, SARIMA has been used in electricity distribution to forecast sales and demand, producing reliable results. When combined with system dashboards, these forecasts help organizations monitor data in real time and make better decisions. These findings show that SARIMA is suitable for forecasting

revenue in systems like fish port operations, where data changes over time and follows seasonal patterns (Mohammed, 2024; Minnaar et al., 2023).

Moreover, SARIMA has been compared with other forecasting models to evaluate its performance. The results show that SARIMA remains a reliable and competitive model, especially when dealing with data that has strong seasonal patterns. Seasonal patterns refer to trends or behaviors in data that repeat at regular intervals, such as monthly or yearly cycles. It can effectively analyze time-based data and support decision-making in different fields. This proves that SARIMA is flexible and useful in predictive systems that require accurate forecasting. Therefore, SARIMA is appropriate for predicting revenue trends in fish port operations, where transaction data follow recurring patterns over time (IEEE, 2024).

Synthesis of the Reviewed Literature and Studies

The reviewed literature shows that many authors agree that digital systems are important in improving port operations. Tijan et al. (2021) and Toro et al. (2024) explained that digital platforms improve communication and coordination among port stakeholders. Zhang et al. (2024) added that technologies like IoT and data analytics help monitor operations more effectively, while Woldesemayat (2025) showed that digital systems improve recording and monitoring of operational activities. Sim et al. (2024) further explained that artificial intelligence can analyze data and support better decision-making. These studies overlap in showing that technology improves port performance and reduces operational problems. These

studies are strong because they are based on modern technologies used in real-world systems. This suggests that digital transformation is necessary for efficient port operations. Among these, the most important idea is the use of data-driven systems to support monitoring and decision-making. This is related to the present study because it also develops a digital system to improve fish port operations.

Many studies also agree that web-based systems are important in managing daily operations. Bueno et al. (2024) and Vega-Quintana et al. (2024) both explained that web-based systems improve data organization and reduce manual work. Ebiesuwa et al. (2023) added that real-time access to data improves coordination among users, while Enaño et al. (2025) showed that these systems improve accuracy and decision-making. Mercurio and Hernandez (2022) further explained that web-based systems support planning and management, while Lapada (2023) added that they improve system performance. These studies share the same idea that web-based systems make work faster, easier, and more organized. These findings are strong because they are supported by practical applications in organizations. This suggests that web-based systems are effective tools for improving productivity and workflow. Among these, the most important idea is real-time access to data for better coordination and decision-making. This is related to the present study because the proposed system will also be web-based for easy access and efficient data management.

In terms of transaction management, the studies show a clear agreement that manual systems are inefficient. Lim and Arbaiy (2022) explained that manual recording leads to errors and delays, while Santos and Reyes (2023) showed that

digital systems improve accuracy and speed through real-time processing. Paulauskas et al. (2021) and Ahasan (2024) emphasized that digital transformation improves operational systems, while Fortinet (2026) highlighted the importance of security through access control. These studies overlap in showing that digital systems are more reliable, accurate, and secure than manual methods. These findings are strong because they focus on real operational problems and provide clear solutions. This suggests that replacing manual systems with digital systems is necessary for improving efficiency. Among these, the most important idea is the improvement of data accuracy and security in transactions. This is related to the present study because it will implement a digital transaction system to ensure accurate and secure data handling.

Another important theme in the literature is stakeholder or user information management. Wang et al. (2022) and Zhang et al. (2023) explained that profiling systems help organize and manage user data. Glöckler et al. (2023) added that identity management systems help secure user access and prevent unauthorized use. Bernardo et al. (2024) emphasized that proper data governance ensures data accuracy and consistency, while Hikmawati et al. (2021) explained that centralized data systems improve data quality. These studies share the idea that proper data management improves system performance and reliability. These studies are strong because they focus on both organization and security of data. This suggests that managing user data properly is important for efficient system operations. Among these, the most important idea is maintaining accurate and secure data for

reliable operations. This is related to the present study because it includes stakeholder profiling features to manage broker and buyer information.

Lastly, the literature shows that predictive analytics is important for planning and decision-making. Sirisha et al. (2022) and WDSI (2022) explained that SARIMA can analyze time-based data and predict trends. Mohammed (2024) and Minnaar et al. (2023) supported this by showing that SARIMA is effective for long-term forecasting. IEEE (2024) also confirmed that SARIMA is reliable for data with seasonal patterns. These studies show that predictive models help organizations prepare for future changes. This suggests that predictive analytics is useful for improving planning and decision-making. Among these, the most important idea is the ability to forecast future trends using past data.

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CHAPTER III

METHODOLOGY

This chapter presents the methodology and step-by-step procedures that were used by the researchers in the development of the Pasacao Fish Port Operations Management with Predictive Analytics System. It discusses the research design, research method, and the system development approach applied in the study.

Research Design

This study employed a developmental research design supported by qualitative data collection. Developmental research focuses on designing, developing, and evaluating systems intended to solve existing problems within a specific environment. The researchers conducted interviews with personnel from the Port Management Unit in Pasacao, Camarines Sur, including the Port Manager and Statistician. These interviews allowed the researchers to gather relevant and reliable information regarding the current operational processes of the port. Through this approach, the researchers ensured that the system being developed is aligned with real-world needs and operational requirements.

The interviews provided detailed insights into the procedures in recording transactions, fee assessment, and report preparation. The researchers were also able to identify several challenges associated with manual record-keeping. These challenges include time-consuming processes, susceptibility to human error, and difficulty in retrieving historical data. The collected information served as the

foundation for identifying both functional and non-functional requirements of the system. As a result, the proposed system was designed to improve efficiency, accuracy, and overall data management within the Port Management Unit.

Research Methodology

This study adopted the Agile Software Development Methodology to guide the development of the proposed system. Agile is an iterative and flexible approach that emphasizes collaboration, continuous feedback, and incremental delivery of system features. It allows developers to divide the system into smaller components that can be developed and improved over time. This approach ensures that the system evolves based on user needs and feedback. It also allows early identification of issues and continuous enhancement of system functionality.

According to Rath et al. (2025), Agile enables development teams to incorporate continuous feedback from stakeholders during development cycles. This improves adaptability and allows the system to respond effectively to changing requirements. Agile also supports faster delivery of functional system features and enables early detection of errors. These advantages make Agile a suitable methodology for system development projects that require flexibility and user involvement. In this study, Agile was used to ensure that the system is continuously improved throughout the development process. The involvement of PMU personnel during development also helped validate system features and usability.

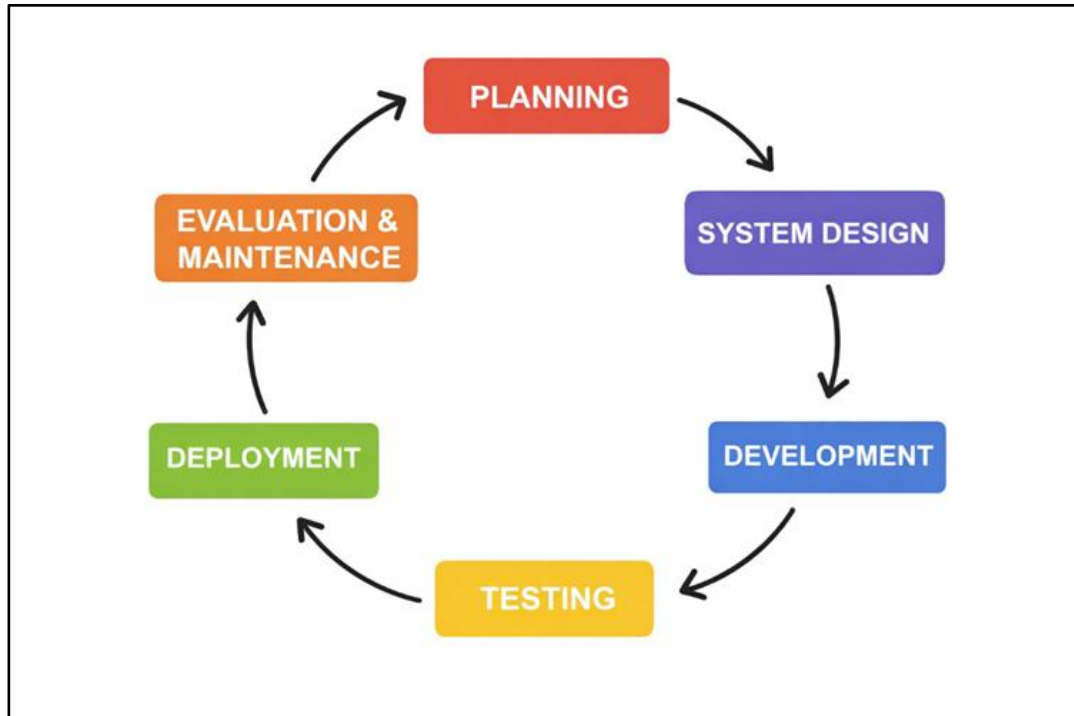


Figure 3.1
Agile Software Development

Phase 1: Planning

This phase focused on defining the project's scope and identifying the operational requirements of the Port Management Unit (PMU) in Pasacao, Camarines Sur. Through this process, the foundation for efficient and systematic development was established, ensuring alignment with the PMU's operational needs while minimizing potential implementation challenges. As part of this phase, the researchers conducted interviews, document analysis, and field observations related to fish port operations. The qualitative data collected provided valuable insights into the existing manual processes, operational limitations, and system requirements. These activities enabled the researchers to clearly understand the

workflow of all transactions and the challenges encountered by personnel in managing records and reports.

In particular, interviews were conducted with the designated Officer-in-Charge and staff responsible for processing transactions. These interviews aimed to obtain detailed information regarding the port's background, scope of operations, transaction procedures, fee computation methods, and reporting practices. The researchers also examined existing logbooks and spreadsheet records to understand how data are currently stored and managed. This comprehensive approach allowed the researchers to identify inefficiencies such as delays in processing, errors in manual computation, and difficulty in retrieving historical records. Furthermore, the absence of a centralized and automated system highlighted the need for a digital solution to improve operational efficiency. These findings formed the basis for the design and development of the core features and functionalities of the proposed system.

Table 3.1
Software Requirements

Software	Minimum	Recommended
Operating System	Windows 10	Windows 11
Browser	Google Chrome, Microsoft Edge	
Programming Languages	HTML, CSS, JavaScript, Python	
Development Tools	XAMPP, VS Code	
Database	MySQL	
Web Hosting	Hostinger	
Internet Connection	10 Mbps	15 Mbps or higher

Table 3.1 presents the software requirements essential for the development and deployment of the Fish Port Operations Management with Predictive Analytics System. To ensure accessibility and compatibility, the system is designed to

operate on modern operating systems, with Windows 10 as the minimum requirement and Windows 11 as the recommended platform. The system supports widely used web browsers such as Google Chrome and Microsoft Edge to provide a stable and consistent user interface. The platform is developed using standard web technologies including HTML, CSS, and JavaScript for the frontend, and PHP for backend processing and system logic. MySQL is used as the database management system to efficiently store and manage transaction records and user data.

The development environment utilizes XAMPP as the local server to facilitate testing and debugging, while Visual Studio Code serves as the primary Integrated Development Environment (IDE) for coding and system development. For system deployment, a reliable web hosting service is required to make the application accessible over the internet and to ensure continuous system availability. Web hosting enables secure data storage, system uptime, and remote access for authorized users, which are critical for the system's real-time operations and data management.

The system architecture is designed to support real-time data recording, retrieval, and report generation. Given the web-based nature of the platform, both web hosting and network connectivity play important roles in overall system performance. A minimum internet connection speed of 10–15 Mbps is required to support smooth system operation, while higher bandwidth is recommended for optimal performance when processing large volumes of transaction data.

Table 3.2 Hardware Requirements

Hardware	Minimum	Recommended
Memory	4 GB RAM	8 GB RAM
Processor	Intel Core i3	Intel Core i5 or higher
Storage	256 GB HDD	512 GB SSD
Network Connectivity	10-15 Mbps	15 Mbps or higher

Table 3.2 outlines the hardware requirements necessary to operate the Fish Port Operations Management with Predictive Analytics System efficiently. These specifications are established to ensure consistent system performance, prevent operational delays, and maintain data integrity during daily transaction processing. The system is designed to handle continuous recording of all transactions, fee computation, and report generation. As such, reliable hardware resources are essential to support smooth and uninterrupted system operation. These requirements ensure that the system can function effectively under regular operational workloads within the Port Management Unit.

The system's memory allocation is designed to support multitasking and the processing of multiple transaction entries without significant delay. A higher memory capacity is particularly beneficial when the system manages large volumes of historical fish landing records and generates reports simultaneously. This allows users to perform tasks such as data encoding, searching, and report viewing efficiently. Adequate memory also improves system responsiveness and reduces processing time. As a result, the recommended memory specification enhances overall system performance and user experience.

Regarding storage, the use of high-speed drives is recommended to facilitate faster data retrieval and improve system reliability. Since the system

continuously stores transaction records, user profiles, and operational data, efficient storage is essential. Faster storage devices, such as solid-state drives (SSD), help reduce loading time and improve database access speed. This is particularly important when retrieving past records and generating reports. Reliable storage also helps ensure the safety and integrity of stored data over time.

To support real-time data recording and retrieval, the system relies on stable network connectivity. A consistent internet connection allows users to access the system, update transaction records, and generate reports without interruption. Although the system operates within a local or web-based environment, network stability is important for maintaining efficient system performance. A minimum connection speed ensures basic functionality, while higher bandwidth improves responsiveness during heavy system usage. Stable connectivity also supports data synchronization and system accessibility for authorized users.

Additionally, the system is designed to be accessible through standard desktop and laptop devices used by Port Management Unit personnel. This ensures that users can conveniently operate the system within their working environment. The system does not require specialized hardware, making it practical and cost-efficient for implementation. The compatibility with commonly available devices enhances usability and accessibility. Overall, these hardware requirements ensure that the system operates efficiently and supports daily port management activities.

The researchers also created a Gantt chart (Figure 3.2) as a guide to track the development of the Fish Port Operations Management with Predictive Analytics System. The Gantt chart served as a fundamental project management tool in organizing and visualizing the entire project timeline and outlining key phases such as planning, system development, testing, and deployment with their corresponding tasks and milestones. This visual representation enabled the researchers to efficiently allocate time and resources, identify task dependencies, and monitor progress throughout the study. It also ensured that important processes, such as database design, system development, debugging, testing, and deployment, were completed in the correct sequence. As the project progressed, the Gantt chart served as a dynamic roadmap that allowed the researchers to compare actual accomplishments with the planned schedule, helping minimize delays and ensure the timely and successful completion of the system.

Phase 2. System Design

In this phase, the researchers identified the necessary tools, technologies, and system requirements for developing the Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao, focusing on transforming user needs into a functional system design that can be presented to and validated by the client. The design process involved converting these requirements into system architecture, data flow, and interface structures, while applying iterative design techniques to refine system components based on expected user interactions, ensuring that the final design is efficient, user-friendly, and aligned with the operational needs of the Port Management Unit (PMU).

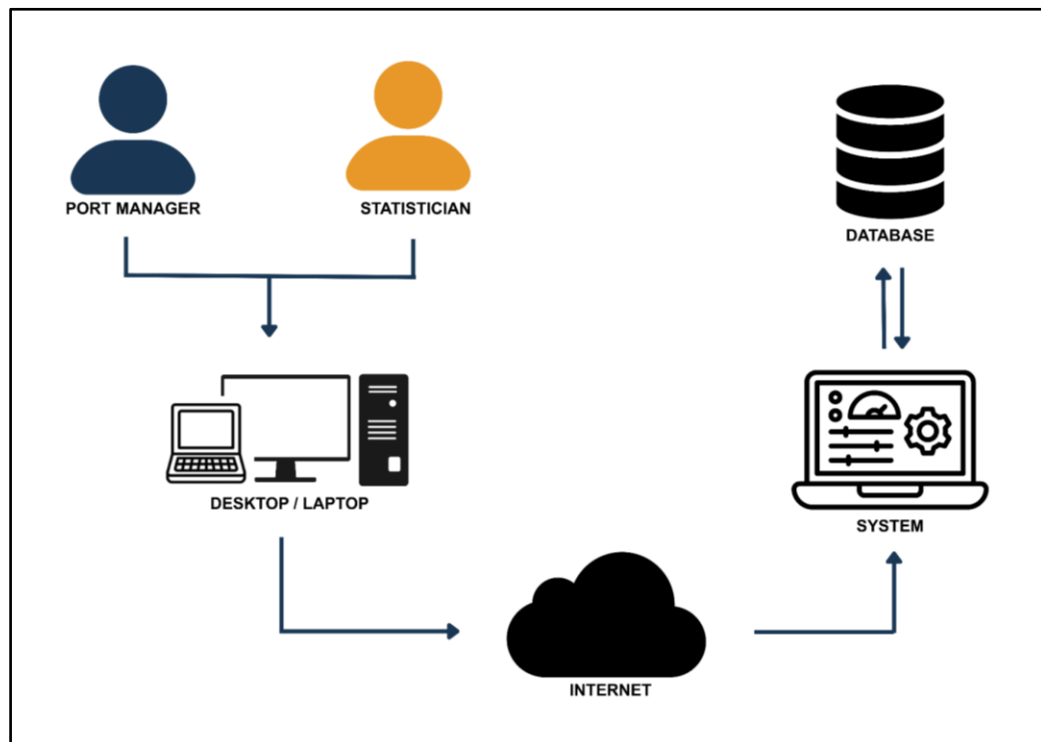


Figure 3.3
System Architecture Design of Web-based Management System with Predictive Analytics for Port Management Unit of Pasacao

Figure 3.3 presents the system architecture used in Phase 2 of the Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao. In this setup, two main users, the Port Manager and the Statistician, interact with the system using internet-enabled devices such as desktop computers and laptops. The communication between users and the system is carried out through the internet. Once connected, the server processes user inputs and controls access based on user roles. The system server communicates with a database where all data is securely stored. This includes transaction records, stakeholders profiles, and reports. The server manages the flow of data between the user interface and the database. It also ensures that data is accurate, updated, and easily retrievable when needed.

This architecture supports real-time data processing and efficient system performance. It allows users to access the system anytime, as long as there is internet connectivity. The design ensures organized data management, fast processing, and reliable system access. Overall, it provides a structured and efficient environment for managing fish port operations and supporting data-driven decision-making.

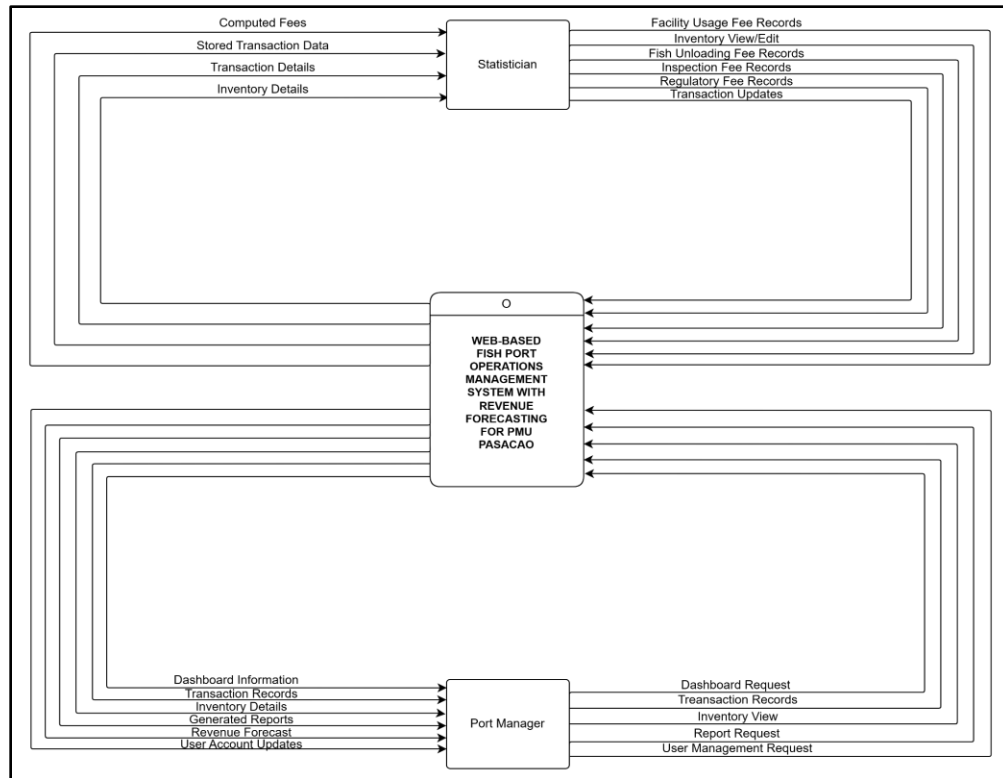


Figure 3.4
Level 0 DFD/Context Diagram of Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao

Figure 3.4 shows the Level 0 Context Diagram of the Web-based Fish Port Operations Management System with Revenue Forecasting, where the system is at the center and interacts with two main users: the Port Manager and the Statistician. The diagram shows how data moves between the users and the system to support daily fish port operations such as recording transactions, managing data, and making decisions. It provides a general view of how the system works with its environment.

At the center of the diagram, the FWeb-based Fish Port Operations Management System with Revenue Forecasting serves as an integrated platform for processing transactions, storing data, generating reports, and performing

predictive analysis. It acts as the core system where all operational data are received, processed, and transformed into meaningful outputs. The system ensures that all information is centralized and easily accessible to authorized users. It also supports real-time processing to improve operational efficiency.

The Statistician is responsible for handling data entry and operational inputs within the system. This includes encoding facility usage fee records, profile view and edits, fish unloading fee records, inspection fee records, regulatory fee records, and transaction updates. In return, the system provides computed fees, stored transaction data, transaction details, and inventory details. This interaction helps minimize manual errors and improves data reliability.

The Port Manager interacts with the system as the primary decision-maker and supervisory user. The manager provides dashboard requests, transaction records, inventory view, report requests, and user management requests. The system responds by dashboard information, transaction records, profiling details, generated reports, revenue forecast and user account updates. This enables the manager to make informed and data-driven decisions.

As a whole, the context diagram illustrates a system that connects operational users and management. It demonstrates how data flows efficiently from input to processing and output. The system ensures accurate recording, organized storage, and meaningful analysis of data. Ultimately, it transforms manual processes into a more efficient and data-driven port management system.

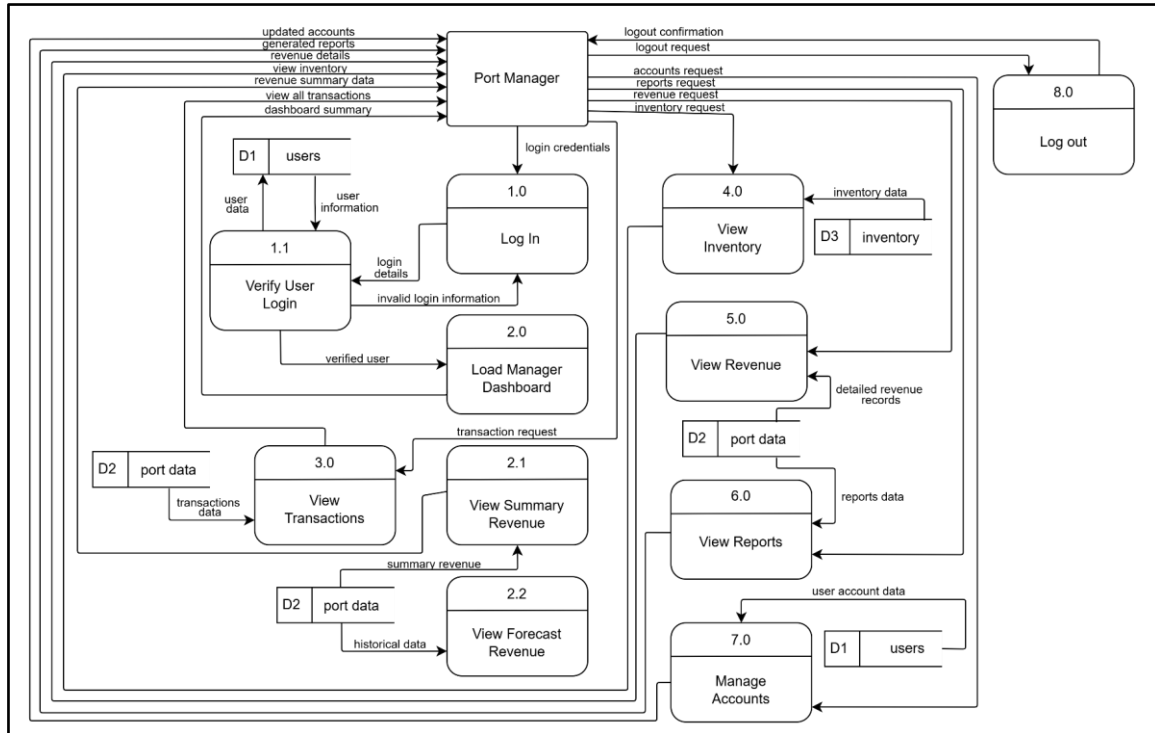


Figure 3.5.1
Part 1 of Level 1 DFD of Fish Port Operation Management with Predictive Analytics System

Figure 3.5.1 presents the Data Flow Diagram (DFD) Level 1 for the Port Manager in the Fish Port Operations Management with Predictive Analytics System. It highlights how the system is mainly used for monitoring port operations and reviewing important information. The system interface includes a sidebar menu that provides access to different features. Through this setup, the Port Manager can easily navigate the system and monitor activities efficiently.

The process begins when the Port Manager logs into the system by entering a username and password. These credentials are processed and validated against the Users data store to ensure that only authorized users can access the system. Once verified, the Port Manager is directed to the dashboard, which serves as the main interface. The dashboard displays important information such as summaries

and revenue forecasts, allowing the Port Manager to quickly monitor the current status of operations. This helps the user get an overview before accessing other features.

Aside from the dashboard, the Port Manager can use the sidebar to access other system features such as transactions, profiling, revenue, reports, and accounts. Each feature retrieves data from the Port Data store and displays it for monitoring and review. The Port Manager can check transaction records, view inventory information, and analyze revenue data without directly encoding information. Reports can also be generated to summarize operational data over time. Overall, the system supports the Port Manager in monitoring activities, reviewing organized data, and making informed decisions.

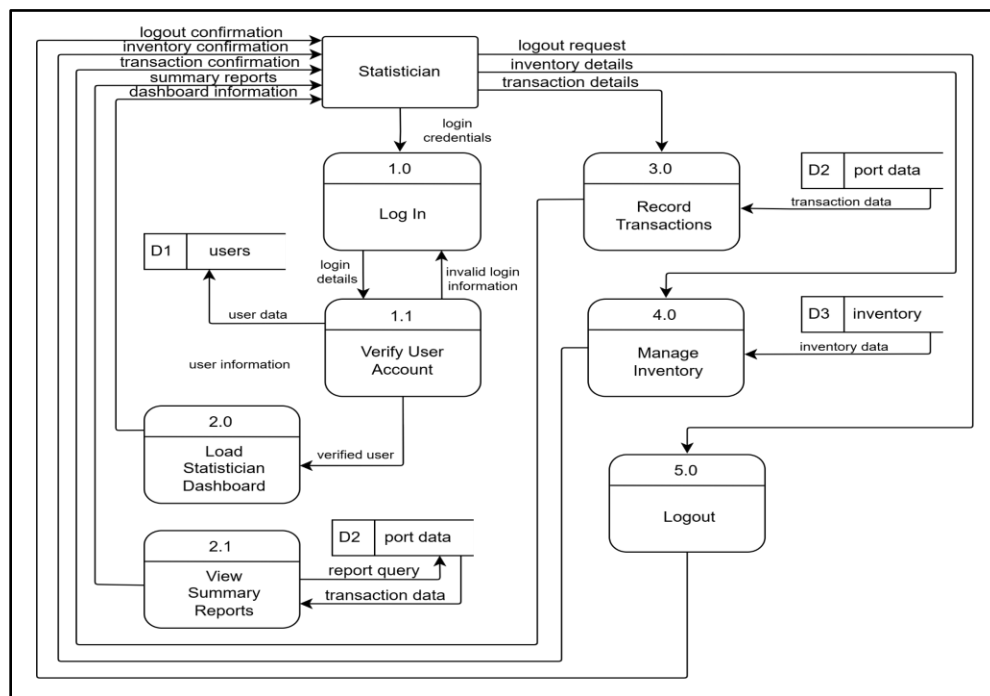


Figure 3.5.2
Part 2 of Level 1 DFD of Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao

Figure 3.5.2 presents the Data Flow Diagram (DFD) Level 1 for the Statistician in the Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao. It highlights how the system is mainly used for encoding, managing, and organizing operational data related to fish port transactions and stakeholder profiling. The system interface provides the Statistician with access to functions such as logging in, loading the dashboard, viewing summary reports, recording transactions, managing inventory, and logging out. Through this setup, the Statistician can efficiently perform daily data management tasks within the system.

The process begins when the Statistician logs into the system by entering login credentials. These credentials are processed and validated through the Verify User Account process using the Users data store to ensure that only authorized users can gain access to the system. If the login details are invalid, the system returns invalid login information. Once verified, the Statistician is granted access to the dashboard, which serves as the main interface for viewing relevant system information and navigating to other functions.

Aside from the dashboard, the Statistician can access other key features such as recording transactions, managing stakeholder profiles, and viewing summary reports. In the Record Transactions process, transaction details are entered and stored in the Port Data store for organized record-keeping and future report generation. In the Manage Inventory process, operational resource inventory is managed and stored in the Inventory database to maintain updated inventory records. The View Summary Reports process retrieves transaction data

from the Port Data store based on report queries and presents summarized reports to the Statistician. The system also provides transaction confirmation, profile confirmation, and logout confirmation as outputs. Overall, the system supports the Statistician in maintaining accurate records, organizing operational data, and generating reports efficiently.

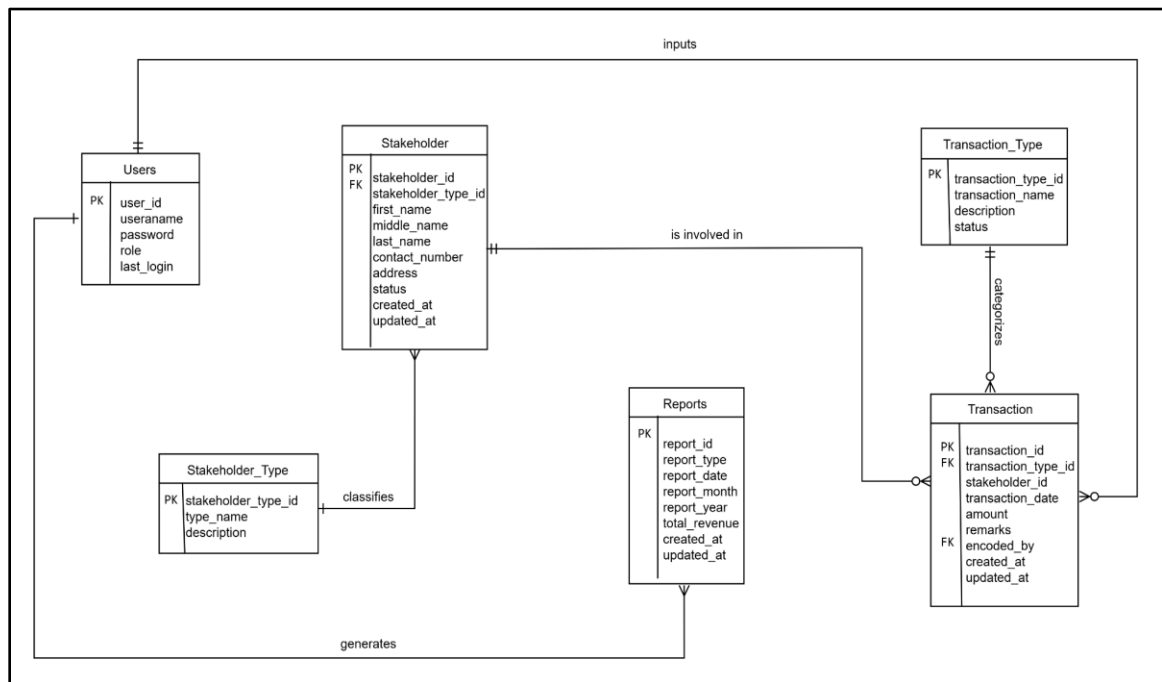


Figure 3.6
Entity Relationship of Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao

Figure 3.6 shows the Entity-Relationship Diagram (ERD) for the Web-based Fish Port Operations Management System with Revenue Forecasting for PMU Pasacao. It presents the flow of structured data and the relationships among the core modules that support stakeholder profiling, transaction recording, revenue monitoring, report generation, and forecasting. The system integrates operational data processing and analytical functions into a single platform that enhances

efficiency, accuracy, and decision-making within the Port Management Unit (PMU). Through this integration, the system ensures that all relevant data is organized, accessible, and usable for both operational and analytical purposes. Each entity is designed to capture essential information that contributes to the overall functionality and usability of the system, ensuring that data is consistent, reliable, and well-structured.

Central to the system is the Users entity, which stores account credentials and role information of individuals accessing the platform. The system is composed of two primary roles: the Port Manager and the Statistician, each with clearly defined responsibilities. These roles determine the level of access and system functionality, ensuring that users interact only with features relevant to their tasks. The Statistician is responsible for recording transactions, managing stakeholder profiles, and producing reports, which are critical to the system's operation. In contrast, the Port Manager is limited to viewing reports, monitoring revenue data, and accessing forecast results, which supports oversight and decision-making. This role-based access ensures controlled interaction with system data while maintaining data integrity, security, and accountability across the platform.

The Stakeholder_Type and Stakeholder entities manage the profiling component of the system. The Stakeholder_Type entity categorizes stakeholders into classifications such as buyer, broker, and renter, providing a structured way to group individuals involved in port operations. This classification allows the system to efficiently organize stakeholder records and support easier filtering and

reporting. The Stakeholder entity stores relevant personal and contact information, including name, contact number, and address, ensuring that accurate records are maintained. By separating classification from profile data, the system avoids redundancy and improves data consistency. This design also supports efficient monitoring and management of stakeholders, which is essential for operational tracking and coordination within the port.

The Transaction_Type and Transaction entities form the core of the system's operational data. The Transaction_Type entity defines the various fee categories collected by the PMU, such as usage fees, unloading fees, and other service-related charges, allowing for clear categorization of revenue sources. This ensures that each transaction is properly classified, making it easier to generate accurate reports and analyses. The Transaction entity records detailed transaction entries, including transaction date, amount, and associated stakeholder when applicable, providing a complete record of financial activities. Each transaction is categorized according to its type and is recorded by a user, ensuring traceability and accountability. This structure enables the system to maintain accurate and consistent revenue data, which is essential for monitoring daily operations and supporting decision-making.

The Reports entity serves as the system's summarization module, providing organized views of collected revenue data. It stores aggregated revenue data in the form of daily, monthly, and annual reports, including the total revenue collected within specific periods. These reports are generated from transaction records, allowing raw data to be transformed into meaningful summaries. By aggregating

transaction data, the system enables users to easily analyze trends, identify patterns, and evaluate overall financial performance. This functionality is particularly important for monitoring operational efficiency and supporting management decisions. The Reports entity thus plays a critical role in bridging raw transaction data and higher-level analysis.

The Forecast entity represents the predictive analytics component of the system, enabling forward-looking insights based on historical data. It stores projected revenue values derived from historical data found in the Reports entity, particularly the total revenue field. Forecasting utilizes these aggregated values to estimate future revenue trends, allowing the PMU to anticipate financial outcomes and plan accordingly. This capability enhances the system's value by supporting strategic decision-making and resource allocation. Since forecasting is based on summarized historical data rather than individual transaction records, its relationship with Reports is conceptual and represented as a derived dependency. This ensures that the forecasting process remains flexible and accurately reflects long-term trends.

The ERD demonstrates a structured and integrated design where raw transaction data is recorded, categorized, and aggregated into reports, which then serve as the foundation for forecasting. The system emphasizes data integrity by ensuring that all entities are properly related and that redundancy is minimized. It also supports role-based access control, ensuring that users interact with the system according to their responsibilities. Additionally, the design promotes efficient data processing by organizing information into logical and manageable

components. By connecting stakeholder profiling, transaction management, reporting, and forecasting, the system provides a comprehensive digital solution for managing port operations. This design also establishes a strong foundation for future enhancements, such as advanced analytics, automated reporting, and improved decision-support tools.

Seasonal Autoregressive Integrated Moving Average (SARIMA)

After conducting Exploratory Data Analysis (EDA), the historical data are explored and visualized to understand their main features, identify patterns, and examine relationships among variables. EDA provides a clear understanding of the dataset, including the number of features, data types, and data distribution. It also helps reveal patterns and relationships between variables, identify errors and outliers that may affect the analysis, highlight the most important features for model building, and support the selection of appropriate modeling techniques for improved results (GeeksforGeeks, 2026).

To further understand how the SARIMA model operates in time-series forecasting, it can be expressed as follows:

$$(1 - \phi_1 B)(1 - \Phi_1 B^s)(1 - B)(1 - B^s)y_t = (1 + \theta_1 B)(1 + \Theta_1 B^s)\varepsilon_t$$

In the equation, y_t represents the actual value of the data at time t , such as the recorded revenue for a specific period. The symbol B refers to the backshift or lag operator, which means using previous values of the data. The parameter ϕ represents the non-seasonal autoregressive component, which uses recent past

values to predict the current value. The parameter Φ represents the seasonal autoregressive component, which uses past values from the same season. The parameter θ represents the non-seasonal moving average component, which uses recent past errors to improve prediction accuracy. The parameter Θ represents the seasonal moving average component, which uses past seasonal errors to refine the forecast. The parameter d indicates the number of times the data is differenced to remove trends and make it stable. The parameter D indicates the number of times seasonal differencing is applied to remove repeating seasonal patterns. The parameter s represents the seasonal period, or how often the pattern repeats, such as monthly or yearly intervals. Lastly, ε_t represents the random error term, which accounts for unpredictable variations that the model cannot explain (GeeksforGeeks, 2025).

NOTES

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**WEB-BASED MANAGEMENT SYSTEM WITH PREDICTIVE ANALYTICS FOR
PORT MANAGEMENT UNIT OF PASACAO**

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May 19, 2026